N 853

Seat No. | | | | | | | |

2025 III 12 1100 – N 853 – SCIENCE AND TECHNOLOGY (72) – PART II (E)

(REVISED COURSE)

Time : 2 Hours (Pages 8) Max. Marks : 40

Note :— (*i*) **All questions are compulsory.**

(*ii*) **The numbers to the right of the questions indicate full marks.**

(*iii*) **In case of MCQs [Q. No. 1 (A)] only the first attempt will be evaluated and will be given credit.**

(*iv*) **Scientifically and technically correct, labeled diagrams should be drawn wherever necessary.**

(*v*) **Each new question should be started on the new page.**

P.T.O.

1. (A) **Choose the *correct* option and write the correct alternative :** 6

(*i*) A molecule of glucose is completely oxidised in aerobic respiration and molecules of H_2O and ..... are produced along with energy.

(A) CO_2

(B) O_2

(C) NaOH

(D) HNO_3

(*ii*) Vinegar contains ..... acid.

(A) acetic

(B) lactic

(C) tartaric

(D) hydrochloric

(*iii*) Any nucleotide of the gene suddenly changes its position that causes a minor change which is nothing but the .....

(A) changes

(B) mutation

(C) disability

(D) translocation

(*iv*) In humans, sperm production occurs in the ..... organ.

(A) vas deferens

(B) ejaculatory duct

(C) testes

(D) urinogenital duct

(*v*) Solar photovoltaic cell converts the solar radiation energy directly into ..... energy.

(A) Chemical

(B) Solar

(C) Mechanical

(D) Electrical

(B) Answer the following questions : 5

(*i*) Find the odd one out :

Fragmentation, Regeneration, Budding, Fertilization.

(*ii*) Write the correct co-relation :

Diabetes : Insulin : : Cancer : .....

(*iii*) State whether true or false :

Study of fossils is an important aspect of the study of evolution.

(*iv*) Give any *two* commercial uses of Biotechnology.

(*v*) Being a fish, I perform respiration with the help of lungs. Who am I ?

2. (A) Give scientific reasons (any *two*) : 4

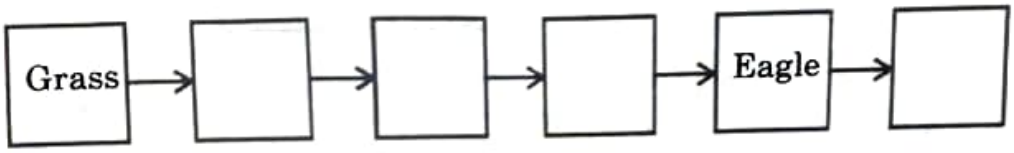
(*i*) Sometimes, higher plants and animals too perform anaerobic respiration.

(*ii*) It is not right to consider the mother responsible, for giving birth to a girl child.

(*iii*) The energy generated from nuclear fuels is not environment friendly.

(B) Answer the following questions (any *three*) : **6**

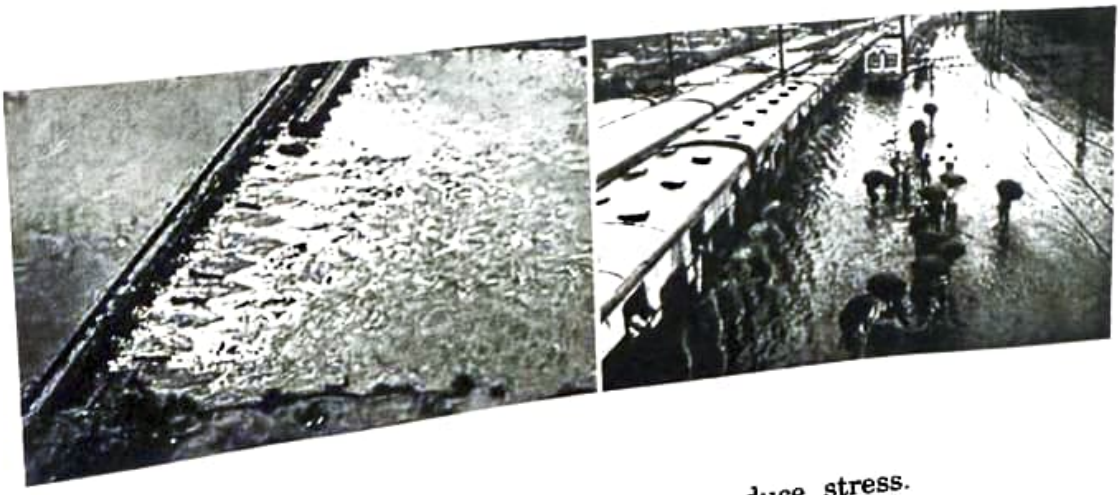
(*i*) Complete the following food chain :



(*ii*) Distinguish between the following :

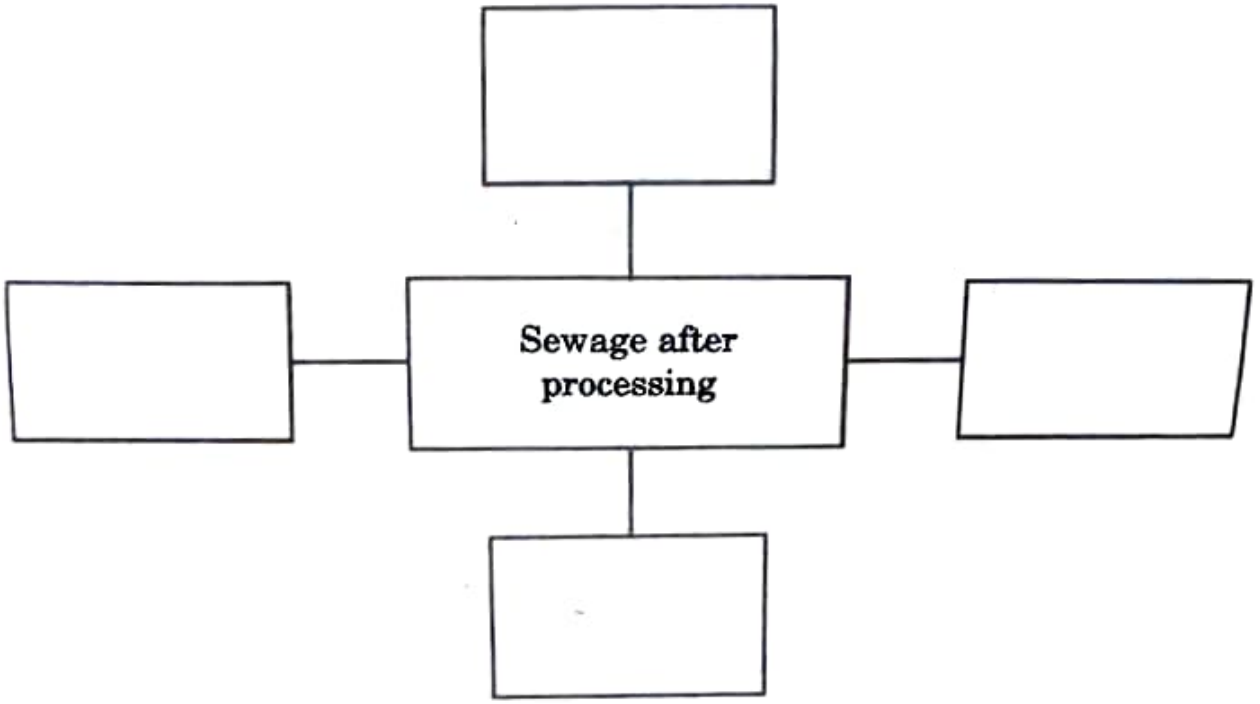
Class Pisces and Class Reptiles

(*iii*) Observe the following picture and write any *two* effects of this disaster :



(*iv*) Write any *four* methods to reduce stress.

(*v*) Complete the following conceptual picture :



3. **Answer the following questions (any *five*) :** **15**

(*i*) Define heredity. Explain the mechanism of hereditary changes.

(*ii*) Write any *four* stages of mitosis and explain any *two* of them.

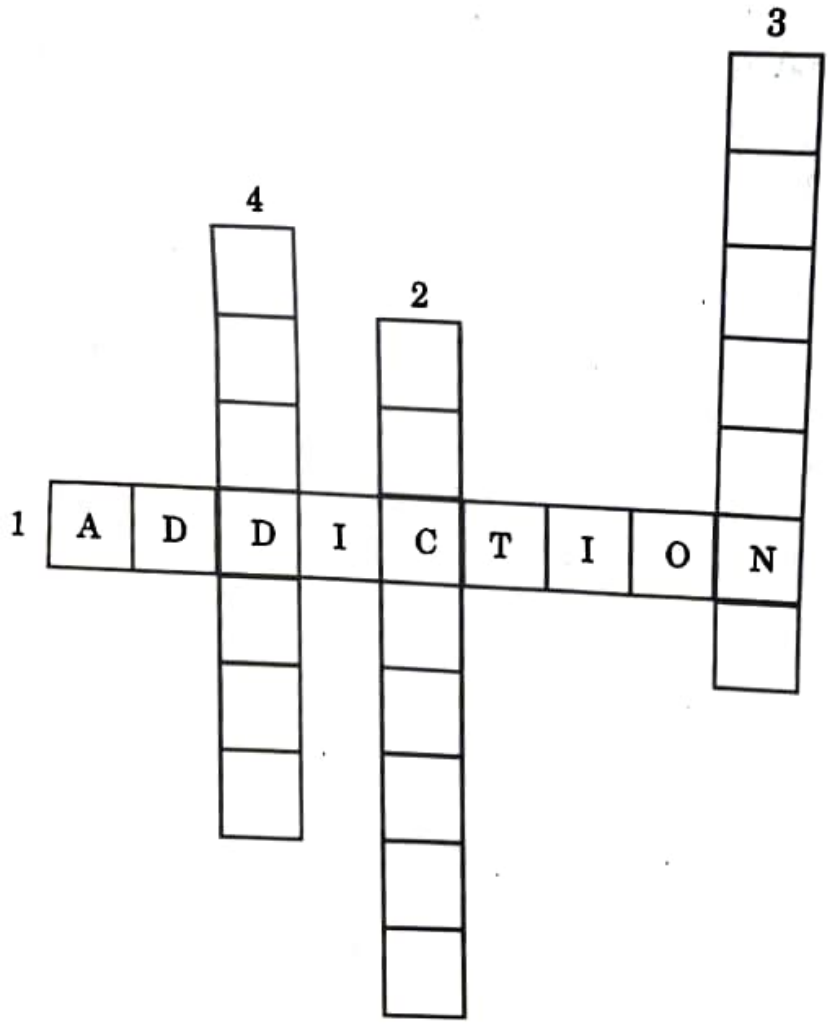
(*iii*) Explain with examples types of asexual reproduction in unicellular organism.

(*iv*) Which fuel is used in thermal power plant ? Write any *two* problems associated with this type of power generation.

(*v*) Write *three* characteristics of phylum-Mollusca.

(vi) What are the constituents of acid rain ? What are its *two* effects on earth's surface ?

(*vii*) Solve the following crossword :



(1) Continuous consumption of alcoholic and tobacco materials.

(2) This app may cause cyber crimes.

(3) A remedy to resolve stress.

(4) Requirement for stress free life.

P.T.O.

(*viii*) Write down the reasons, effects and remedial measures taken for any *one* disaster seen or heard by you.

4. Answer the following questions (any *one*) : 5

(*i*) What is biodiversity ? Write any *four* steps to conserve biodiversity.

(*ii*) What is stem cell ? Write any *four* uses of stem cell.